Name: Lobo, Siantil Noelyn M.
Course & Section: BSIT WMT 2-2

W1-Quiz1 Activity: Introduction to Algorithms

A. Discuss Briefly

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?
An algorithm is just a list of steps to do something. In daily life, we use algorithms like following a recipe, brushing teeth, or riding a jeep. You follow the steps one by one to reach the goal.

2. Why is it important for an IT student to learn algorithms?
Because programming is all about steps. If we know how to make good algorithms, we can make programs that are faster and easier to understand. It helps us become better problem solvers in IT.

B. Step-by-step Algorithm / Pseudocode

1. Code:

```
for (let i = 1; i < 10; i *= 2) {
  console.log(i);
}
```

Algorithm:
1. Start with “i = 1”
2. If `i` is less than 10, print “i”
3. Multiply “i” by 2
4. Repeat step 2 until “i” is no longer less than 10
5. Stop

2. Code:

```
for (let i = 0; i < 10; i++) {
  console.log(i);
}
```

Algorithm:
1. Start with “i = 0”
2. If “i” is less than 10, print “i”
3. Add 1 to “i”
4. Repeat step 2 until “i” is no longer less than 10
5. Stop

C. My Observations

Output of Code 1: “1, 2, 4, 8”
Output of Code 2: “0, 1, 2, 3, 4, 5, 6, 7, 8, 9”

1. First code:
The numbers double. It starts at 1 then 2, 4, 8 and stops.

2. Second code:
The numbers just count up by 1 from 0 to 9.

3. Main difference:
First loop multiplies by 2 so it gets big fast. Second loop just adds 1 so it goes slow.

4. How steps affect output?
The steps in the loop decide what numbers come out. If you change how “i” changes, the whole result changes too.